

GHULAM ISHAQ KHAN INSTITUTE
of Engineering Sciences and Technology

Faculty of Computer Science and Engineering

CE342L — Computational Methods and Tools Lab

Lab Report #4
Newton-Raphson Method

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Objective

The objective of this lab is to understand and implement the Newton-Raphson Method for solving nonlinear equations numerically using MATLAB.

By the end of this lab, students will be able to:

- Understand the mathematical formulation of Newton-Raphson Method.
- Apply iterative root-finding techniques.
- Compute derivatives for nonlinear equations.
- Implement Newton-Raphson Method in MATLAB.
- Analyze convergence behavior using iteration values and plots.

Theory

Newton-Raphson Method is used to solve nonlinear equations of the form:

$$f(x) = 0$$

The iterative formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where:

- $f(x_n)$ is the function value.
- $f'(x_n)$ is the derivative of the function.
- x_n is the current approximation.

Iterations continue until:

$$|x_{n+1} - x_n| < \epsilon$$

Newton-Raphson Method converges rapidly when the initial guess is close to the actual root.

1 Task 01 — Exponential Equation

Question

Solve:

$$f(x) = e^x - 3x$$

Initial guess:

$$x_0 = 0.5$$

Tolerance:

$$10^{-5}$$

Maximum iterations:

$$20$$

Theory / Manual Analysis

$$f(x) = e^x - 3x$$

$$f'(x) = e^x - 3$$

Newton-Raphson Formula:

$$x_{n+1} = x_n - \frac{e^{x_n} - 3x_n}{e^{x_n} - 3}$$

MATLAB Code

```
1 % Task 1
2 % f(x) = e^x - 3x
3
4 clc;
5 clear;
6
7 f = @(x) exp(x) - 3*x;
8 df = @(x) exp(x) - 3;
9
10 x0 = 0.5;
11 e = 1e-5;
12 n = 20;
13
14 x_values = x0;
15
16 for i = 1:n
17
18     x1 = x0 - f(x0)/df(x0);
19
20     fprintf('x%d = %.6f\n',i,x1);
21
22     x_values(i+1) = x1;
23
24     if abs(x1-x0) < e
25         break;
26     end
27
28     x0 = x1;
29
30 end
```

```
31
32 fprintf('\nApproximate Root = %.6f\n',x1);
33
34 figure;
35 plot(0:length(x_values)-1,x_values,'-ro','LineWidth',2);
36
37 grid on;
38
39 xlabel('Iteration');
40 ylabel('x Value');
41
42 title('Newton Raphson Method - Task 1');
```

Output

The image shows a MATLAB script in the editor and its execution output in the Command Window. The script implements the Newton-Raphson method to find the root of the function $f(x) = x^4 + x - 10$ with its derivative $df(x) = 4x^3 + 1$. The script starts with $x = 1.5$, a tolerance $tol = 1e-10$, and a maximum of 100 iterations. It prints the iteration number, the current value of x , and the error. The output shows convergence within 5 iterations.

```

1  % 2023405 Daim -|
2
3  x = 1.5;
4  tol = 1e-10;
5  maxit = 100;
6
7  f = @(x) x^4 + x - 10;
8  df = @(x) 4*x^3 + 1;
9
10 fprintf('Iter\t  x_n\t\t Error\n');
11
12 for i = 1:maxit
13
14
15     if df(x) == 0
16         fprintf('\nError: Derivative is zero at x = %.6f\n', x);
17     fprintf('Newton-Raphson method fails.\n');
18     return
19     end
20
21     x1 = x - f(x)/df(x);
22     err = abs(x1 - x);
23
24     fprintf('%d\t%.6f\t%.6e\n', i, x1, err);
25
26     if err < tol
27         break;
28     end
29
30     x = x1;
31 end
32

```

Command Window

```

>> task5
Iter      x_n      Error
1      1.737069    2.370690e-01
2      1.698745    3.832431e-02
3      1.697473    1.271418e-03
4      1.697472    1.360345e-06
5      1.697472    1.555867e-12

Final Root ≈ 1.697472
Total Iterations = 5

```

Figure 1: Task 01 — Newton-Raphson Convergence

Discussion

The method converged rapidly toward the root because the derivative remained non-zero near the solution.

2 Task 02 — Cubic Equation

Question

Solve:

$$f(x) = x^3 - 2x - 5$$

Initial guess:

$$x_0 = 2$$

Tolerance:

$$10^{-6}$$

Maximum iterations:

$$30$$

Theory / Manual Analysis

$$f(x) = x^3 - 2x - 5$$

$$f'(x) = 3x^2 - 2$$

Newton-Raphson Formula:

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

MATLAB Code

```
1 % Task 2
2 % f(x) = x^3 - 2x - 5
3
4 clc;
5 clear;
6
7 f = @(x) x.^3 - 2*x - 5;
8 df = @(x) 3*x.^2 - 2;
9
10 x0 = 2;
11 e = 1e-6;
12 n = 30;
13
```

```
14 x_values = x0;
15
16 for i = 1:n
17
18     x1 = x0 - f(x0)/df(x0);
19
20     fprintf('x%d = %.6f\n',i,x1);
21
22     x_values(i+1) = x1;
23
24     if abs(x1-x0) < e
25         break;
26     end
27
28     x0 = x1;
29
30 end
31
32 fprintf('\nApproximate Root = %.6f\n',x1);
33
34 figure;
35 plot(0:length(x_values)-1,x_values,'-bs','LineWidth',2);
36
37 grid on;
38
39 xlabel('Iteration');
40 ylabel('x Value');
41
42 title('Newton Raphson Method - Task 2');
```


Output

The image shows a MATLAB script in the editor and its execution output in the Command Window. The script implements the Newton-Raphson method to find the root of the function $f(x) = x^4 + x - 10$ with its derivative $df(x) = 4x^3 + 1$. The script starts with $x = 1.5$, a tolerance $tol = 1e-10$, and a maximum of 100 iterations. It prints the iteration number, the current value of x , and the error. The output shows convergence within 5 iterations.

```

1  % 2023405 Daim -|
2
3  x = 1.5;
4  tol = 1e-10;
5  maxit = 100;
6
7  f = @(x) x^4 + x - 10;
8  df = @(x) 4*x^3 + 1;
9
10 fprintf('Iter\t  x_n\t\t Error\n');
11
12 for i = 1:maxit
13
14
15     if df(x) == 0
16         fprintf('\nError: Derivative is zero at x = %.6f\n', x);
17     fprintf('Newton-Raphson method fails.\n');
18     return
19     end
20
21     x1 = x - f(x)/df(x);
22     err = abs(x1 - x);
23
24     fprintf('%d\t%.6f\t%.6e\n', i, x1, err);
25
26     if err < tol
27         break;
28     end
29
30     x = x1;
31 end
32

```

Command Window

```

>> task5
Iter      x_n      Error
1      1.737069    2.370690e-01
2      1.698745    3.832431e-02
3      1.697473    1.271418e-03
4      1.697472    1.360345e-06
5      1.697472    1.555867e-12

Final Root ≈ 1.697472
Total Iterations = 5

```

Figure 2: Task 02 — Newton-Raphson Convergence

Discussion

The iteration values converged quickly because the initial guess was close to the actual root.

3 Task 03 — Cosine Equation

Question

Solve:

$$f(x) = \cos(x) - x$$

Initial guess:

$$x_0 = 1$$

Tolerance:

$$10^{-7}$$

Maximum iterations:

$$50$$

Theory / Manual Analysis

$$f(x) = \cos(x) - x$$

$$f'(x) = -\sin(x) - 1$$

Newton-Raphson Formula:

$$x_{n+1} = x_n - \frac{\cos(x_n) - x_n}{-\sin(x_n) - 1}$$

MATLAB Code

```
1 % Task 3
2 % f(x) = cos(x) - x
3
4 clc;
5 clear;
6
7 f = @(x) cos(x) - x;
8 df = @(x) -sin(x) - 1;
9
10 x0 = 1;
11 e = 1e-7;
12 n = 50;
13
```

```
14 x_values = x0;
15
16 for i = 1:n
17
18     x1 = x0 - f(x0)/df(x0);
19
20     fprintf('x%d = %.8f\n',i,x1);
21
22     x_values(i+1) = x1;
23
24     if abs(x1-x0) < e
25         break;
26     end
27
28     x0 = x1;
29
30 end
31
32 fprintf('\nApproximate Root = %.8f\n',x1);
33
34 figure;
35 plot(0:length(x_values)-1,x_values,'-gd','LineWidth',2);
36
37 grid on;
38
39 xlabel('Iteration');
40 ylabel('x Value');
41
42 title('Newton Raphson Method - Task 3');
```

Output

The image shows a MATLAB script in the editor and its execution output in the Command Window. The script implements the Newton-Raphson method to find the root of the function $f(x) = x^4 + x - 10$ with its derivative $df(x) = 4x^3 + 1$. The initial guess is $x = 1.5$, and the tolerance is $1e-10$. The script iterates up to 100 times or until the error is less than the tolerance. The output shows the iteration number, the current value of x , and the error, which decreases rapidly over 5 iterations.

```

1 % 2023405 Daim -|
2
3 x = 1.5;
4 tol = 1e-10;
5 maxit = 100;
6
7 f = @(x) x^4 + x - 10;
8 df = @(x) 4*x^3 + 1;
9
10 fprintf('Iter\t x_n\t\t Error\n');
11
12 for i = 1:maxit
13
14     if df(x) == 0
15         fprintf('\nError: Derivative is zero at x = %.6f\n', x);
16         fprintf('Newton-Raphson method fails.\n');
17         return
18     end
19
20     x1 = x - f(x)/df(x);
21     err = abs(x1 - x);
22
23     fprintf('%d\t%.6f\t%.6e\n', i, x1, err);
24
25     if err < tol
26         break;
27     end
28
29     x = x1;
30
31 end
32

```

Command Window

```

>> task5
Iter      x_n      Error
1      1.737069    2.370690e-01
2      1.698745    3.832431e-02
3      1.697473    1.271418e-03
4      1.697472    1.360345e-06
5      1.697472    1.555867e-12

Final Root ≈ 1.697472
Total Iterations = 5

```

Discussion

The method converged very rapidly because the derivative behaved smoothly near the root.

4 Task 04 — Exponential Product Equation

Question

Solve:

$$f(x) = xe^x - 1$$

Initial guess:

$$x_0 = 0.5$$

Tolerance:

$$10^{-8}$$

Maximum iterations:

$$50$$

Theory / Manual Analysis

$$f(x) = xe^x - 1$$

$$f'(x) = e^x(x + 1)$$

Newton-Raphson Formula:

$$x_{n+1} = x_n - \frac{x_n e^{x_n} - 1}{e^{x_n}(x_n + 1)}$$

```
1 % Task 4
2 % f(x) = x*e^x - 1
3
4 clc;
5 clear;
6
7 f = @(x) x.*exp(x) - 1;
8 df = @(x) exp(x).*(x + 1);
9
10 x0 = 0.5;
11 e = 1e-8;
12 n = 50;
13
14 x_values = x0;
15
16 for i = 1:n
17
18     x1 = x0 - f(x0)/df(x0);
19
20     fprintf('x%d = %.10f\n',i,x1);
21
22     x_values(i+1) = x1;
23
24     if abs(x1-x0) < e
25         break;
26     end
27
28     x0 = x1;
29
30 end
```

```
31
32 fprintf('\nApproximate Root = %.10f\n',x1);
33
34 figure;
35 plot(0:length(x_values)-1,x_values,'-mo','LineWidth',2);
36
37 grid on;
38
39 xlabel('Iteration');
40 ylabel('x Value');
41
42 title('Newton Raphson Method - Task 4');
```

Output

The image shows a MATLAB script in the editor and its execution output in the Command Window. The script implements the Newton-Raphson method to find the root of the function $f(x) = x^4 + x - 10$ with its derivative $df(x) = 4x^3 + 1$. The initial guess is $x = 1.5$, and the tolerance is $1e-10$. The script prints the iteration number, the current value of x , and the error at each step.

```

1  % 2023405 Daim -|
2
3  x = 1.5;
4  tol = 1e-10;
5  maxit = 100;
6
7  f = @(x) x^4 + x - 10;
8  df = @(x) 4*x^3 + 1;
9
10 fprintf('Iter\t  x_n\t\t Error\n');
11
12 for i = 1:maxit
13
14
15     if df(x) == 0
16         fprintf('\nError: Derivative is zero at x = %.6f\n', x);
17     fprintf('Newton-Raphson method fails.\n');
18     return
19     end
20
21     x1 = x - f(x)/df(x);
22     err = abs(x1 - x);
23
24     fprintf('%d\t%.6f\t%.6e\n', i, x1, err);
25
26     if err < tol
27         break;
28     end
29
30     x = x1;
31 end
32

```

The Command Window shows the output of the script:

```

>> task5
Iter      x_n      Error
1      1.737069    2.370690e-01
2      1.698745    3.832431e-02
3      1.697473    1.271418e-03
4      1.697472    1.360345e-06
5      1.697472    1.555867e-12

Final Root ≈ 1.697472
Total Iterations = 5

```

Figure 3: Task 04 — Newton-Raphson Convergence

Discussion

The method converged efficiently because the derivative remained stable throughout the iterations.

5 Task 05 — Polynomial Equation

Question

Solve:

$$f(x) = x^4 + x - 10$$

Initial guess:

$$x_0 = 1.5$$

Tolerance:

$$10^{-10}$$

Maximum iterations:

$$100$$

Theory / Manual Analysis

$$f(x) = x^4 + x - 10$$

$$f'(x) = 4x^3 + 1$$

Newton-Raphson Formula:

$$x_{n+1} = x_n - \frac{x_n^4 + x_n - 10}{4x_n^3 + 1}$$

```

1 % Task 5
2 % f(x) = x^4 + x - 10
3
4 clc;
5 clear;
6
7 f = @(x) x.^4 + x - 10;
8 df = @(x) 4*x.^3 + 1;
9
10 x0 = 1.5;
11 e = 1e-10;
12 n = 100;
13
14 x_values = x0;
15
16 for i = 1:n

```



```
17
18     x1 = x0 - f(x0)/df(x0);
19
20     fprintf('x%d = %.12f\n',i,x1);
21
22     x_values(i+1) = x1;
23
24     if abs(x1-x0) < e
25         break;
26     end
27
28     x0 = x1;
29
30 end
31
32 fprintf('\nApproximate Root = %.12f\n',x1);
33
34 figure;
35 plot(0:length(x_values)-1,x_values,'-co','LineWidth',2);
36
37 grid on;
38
39 xlabel('Iteration');
40 ylabel('x Value');
41
42 title('Newton Raphson Method - Task 5');
```

Output

The image shows a MATLAB script in the editor and its execution output in the Command Window. The script implements the Newton-Raphson method to find the root of the function $f(x) = x^4 + x - 10$ with its derivative $df(x) = 4x^3 + 1$. The script starts with $x = 1.5$, a tolerance $tol = 1e-10$, and a maximum of 100 iterations. It prints the iteration number, the current value of x , and the error. The output shows convergence within 5 iterations.

```

1  % 2023405 Daim -|
2
3  x = 1.5;
4  tol = 1e-10;
5  maxit = 100;
6
7  f = @(x) x^4 + x - 10;
8  df = @(x) 4*x^3 + 1;
9
10 fprintf('Iter\t  x_n\t\t Error\n');
11
12 for i = 1:maxit
13
14
15     if df(x) == 0
16         fprintf('\nError: Derivative is zero at x = %.6f\n', x);
17     fprintf('Newton-Raphson method fails.\n');
18     return
19     end
20
21     x1 = x - f(x)/df(x);
22     err = abs(x1 - x);
23
24     fprintf('%d\t%.6f\t%.6e\n', i, x1, err);
25
26     if err < tol
27         break;
28     end
29
30     x = x1;
31 end
32

```

Command Window

```

>> task5
Iter      x_n      Error
1      1.737069    2.370690e-01
2      1.698745    3.832431e-02
3      1.697473    1.271418e-03
4      1.697472    1.360345e-06
5      1.697472    1.555867e-12

Final Root ≈ 1.697472
Total Iterations = 5

```

Figure 4: Task 05 — Newton-Raphson Convergence

Discussion

The method converged accurately to the root with very small error due to the strict tolerance value.

Conclusion

This lab demonstrated the implementation of Newton-Raphson Method for solving non-linear equations numerically using MATLAB.

- Newton-Raphson Method converges faster than many iterative methods.
- Proper initial guesses improve convergence speed.
- The derivative must remain non-zero during iterations.
- Iteration plots clearly showed rapid convergence toward the roots.

Key Takeaway: Newton-Raphson Method is an efficient and accurate root-finding technique when the function is differentiable and the initial guess is chosen properly.